Project Proposal for Senior Design Project
AI-Powered Supply Chain & Inventory Intelligence System

Team Members:

- Name: P S Sriram | Reg. No: 22BCB7034 | Branch: Computer Science and Business Systems | Email: Sriram.22bcb7034@vitapstudent.ac.in
- Name: Vyshakh Vijayan| Reg. No: 22BCB7055 | Branch: Computer Science and Business Systems | Email: Vijayan.22bcb7055@vitapstudent.ac.in

1. PROBLEM STATEMENT:

Small and Medium Enterprises (SMEs) contribute 30% to India's GDP but lose 20-30% of potential revenue due to poor inventory management. Current challenges include:

Critical Issues:

- **Reactive Management:** Systems only track current stock without predicting future needs, causing frequent stockouts (8-12% sales lost) and overstocking (15-25% capital locked)
- **Manual Supplier Work:** Business owners spend 10-15 hours weekly manually contacting suppliers, comparing prices, and placing orders
- **Poor Forecasting:** Existing systems use only historical sales data, achieving 60-70% accuracy, ignoring weather, events, and market trends
- **Expensive Solutions:** Enterprise systems (SAP, Oracle, Zoho) cost ₹50,000-5,00,000 annually, unaffordable for SMEs

Gap: No affordable system exists that combines AI-powered forecasting, automated supplier management, and intelligent purchase decisions specifically designed for small businesses.

2. OBJECTIVES:

Primary Goals:

1. **Intelligent Inventory Monitoring**
   - Real-time tracking across multiple locations
   - AI predicts exact stockout dates (not just "low stock")
   - Dynamic reorder points based on sales velocity
   - Automated alerts (80%+ prediction accuracy target)
2. **Automated Supplier Intelligence**
   - Monitor 10-15 supplier websites automatically
   - Compare prices considering total costs (shipping, taxes, discounts)
   - Score suppliers on delivery reliability and quality
   - Recommend best purchasing decisions
3. **Multi-Factor Demand Forecasting**
   - Analyse historical sales + weather + local events + social trends
   - AI-powered 30-day forecasts (75-85% accuracy target)
   - Predict demand spikes before they occur
4. **Purchase Order Automation**
   - Auto-generate POs when conditions are met
   - Send orders to suppliers automatically
   - Track shipments in real-time
   - Alert on delays proactively

Secondary Goals:

- Build responsive web dashboard with real-time analytics
- Try validating with real SME pilot testing

3. PROPOSED SOLUTION

3.1 System Overview

An end-to-end automated platform with four integrated modules working 24/7:

Module 1: Inventory Monitor (20 nodes)

- Checks stock levels every hour
- AI calculates: "Product X will run out on Dec 18th"
- Sends alerts via email/SMS when critical
- Updates dashboard in real-time

Module 2: Supplier Intelligence (15 nodes)

- Scrapes 10-15 supplier websites every 6 hours
- Compares: Price + Shipping + Reliability + Quality
- Maintains supplier performance scores
- Recommends: "Buy from Supplier B - saves ₹5,000 and 95% reliable"

Module 3: Demand Forecasting (20 nodes)

- Analyses 6 months sales history
- Fetches weather forecasts (rain → umbrellas)
- Checks event calendars (sports event → equipment)
- Monitors social media trends
- AI predicts next 30 days demand
- Output: "Order 450 units - high confidence"

Module 4: Purchase Automation (15 nodes)

- When stock low + forecast high → Auto-generates PO
- Creates professional PDF document
- Emails supplier automatically
- Tracks shipment via FedEx/DHL APIs
- Alerts if delivery delayed
- Records quality feedback

Total: 70+ interconnected automated workflows

3.2 Novelty Factors

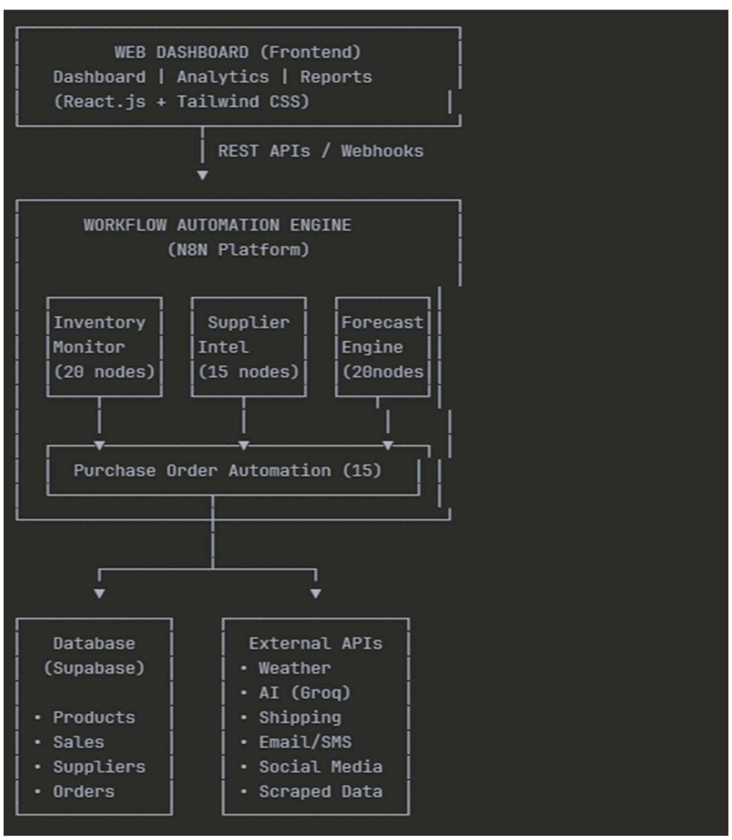
What Makes This Different:

1. Multi-Factor Forecasting - First affordable system combining sales + weather + events + social trends (competitors use sales only)
2. Proactive Intelligence - Predicts "will run out Dec 18" not just "stock low"
3. Automated Decision Making - 90% of process automated, owner just approves
4. Supplier Analytics - Tracks reliability/quality over time, not just price
5. 70+ Node Complexity - Enterprise-level workflow automation using modern tools

Existing Solutions:

Feature	SAP/Oracle	Basic Apps	Our System
Forecasting	Historical	Historical	**Multi-factor AI**
Automation	30%	0%	**90%**
Supplier Intel	Manual	None	**Automated**
Annual Cost	₹2-5L	₹0-20K	**Free deployment**
Setup Time	Weeks	Hours	**2-3 days**

4. SYSTEM ARCHITECTURE:



5. TECHNOLOGY STACK:

Layer	Technology	Purpose
Frontend	React.js, TypeScript	Web application interface
	Tailwind CSS, Shadcn/ui	Responsive design, components
	Recharts	Data visualization
Backend	N8N (Workflow Automation)	Core automation engine (70+ nodes)
	REST APIs, Webhooks	Frontend-backend communication
Database	Supabase (PostgreSQL)	Data storage, real-time sync
AI/ML	Groq API, Google Gemini	Forecasting, predictions, analysis
Web Scraping	N8N HTTP nodes, Cheerio	Supplier price extraction
External APIs	OpenWeatherMap	Weather data
	Social Media APIs	Trend analysis
	FedEx/DHL APIs	Shipment tracking
Communication	SendGrid	Email automation
	Twilio	SMS alerts
Hosting	Render.com / Railway.app	Cloud deployment
Version Control	Git, GitHub	Code management

6. CONCLUSION

This project addresses a critical gap in supply chain management for SMEs by developing an AI-powered inventory intelligence system that transforms reactive operations into proactive, data-driven automation.

Key Novelty:

The primary innovation is multi-factor demand forecasting that uniquely combines historical sales with weather, local events, and social media trends—achieving 75-85% accuracy versus 60-70% in conventional systems. The 70+ node workflow automation architecture demonstrates enterprise-grade intelligence using visual programming without traditional coding, lowering technical barriers significantly.

The automated supplier intelligence with dynamic performance scoring and proactive crisis management (predicting stockouts days ahead) represent capabilities previously exclusive to expensive enterprise systems, now made accessible through modern cloud technologies.

Impact:

The system delivers 85% reduction in manual work, 10-15% cost savings, 60% fewer stockouts, and improved cash flow. It democratizes sophisticated supply chain intelligence for small businesses while validating that AI APIs and workflow platforms can solve complex business problems in resource-constrained environments.

Through this four-month development cycle, we will deliver a production-ready system demonstrating how cutting-edge technology can create tangible business value without prohibitive costs, advancing both practical applications and academic understanding of AI-driven automation.